

CX4240 Spring 2025

Probability and Statistics Revisit

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Office Hours



Friday: 3:00-4:00pm, [Online Session](#)

Office Hours



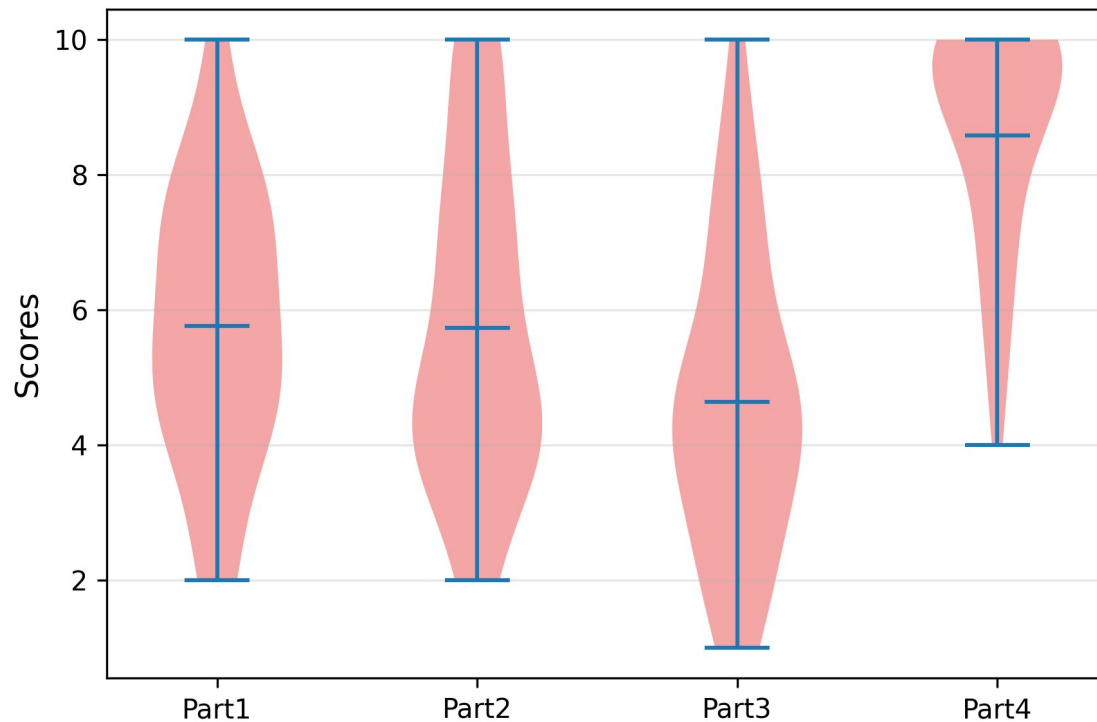
Monday: 2:00-3:00pm, *Coda 2nd floor*. Changhao Li

Thursday: 3:00-4:00pm, *Coda 2nd floor*. Chenxiao Gao

Basic / Prerequisites

- Probability
 - Distributions, densities, marginalization, conditioning
- Statistics
 - Mean, variance, maximum likelihood estimation
- Linear Algebra and Optimization
 - Vector, matrix, multiplication, inversion, eigen-value decomposition
- Coding Skills
 - Pytorch and/or JAX

Statistics of Background Test



Probability and Statistics

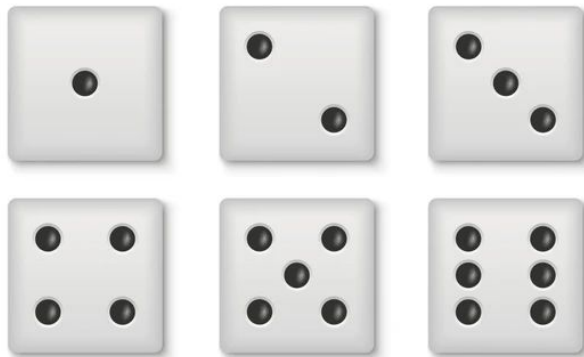
Revist

Basic Probability Concepts

- A **sample space S** is the set of all possible outcomes of a conceptual or physical, repeatable experiment. (S can be finite or infinite.)

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(1 2 3 4 5 6)



Basic Probability Concepts

- A **sample space S** is the set of all possible outcomes of a conceptual or physical, repeatable experiment. (S can be finite or infinite.)
 - E.g., S may be the set of all possible outcomes of a dice roll: S
(1 2 3 4 5 6)
 - E.g., S may be the set of all possible nucleotides of a DNA site: S
(A C G T)
- An **Event A** is any subset of S
 - Seeing "1" or "6" in a dice roll; observing a "G" at a site

Discrete Probability Distribution

- A probability distribution P defined on a discrete sample space S is an assignment of a non-negative real number $P(s)$ to each sample $s \in S$:
 - Probability Mass Function (PMF): $p_x(x_i) = P[X = x_i]$
 - Properties: $p_x(x_i) \geq 0$ and $\sum_i p_x(x_i) = 1$

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- Examples:
 - Bernoulli Distribution:

$$\begin{cases} 1 - p & \text{for } x = 0 \\ p & \text{for } x = 1 \end{cases}$$

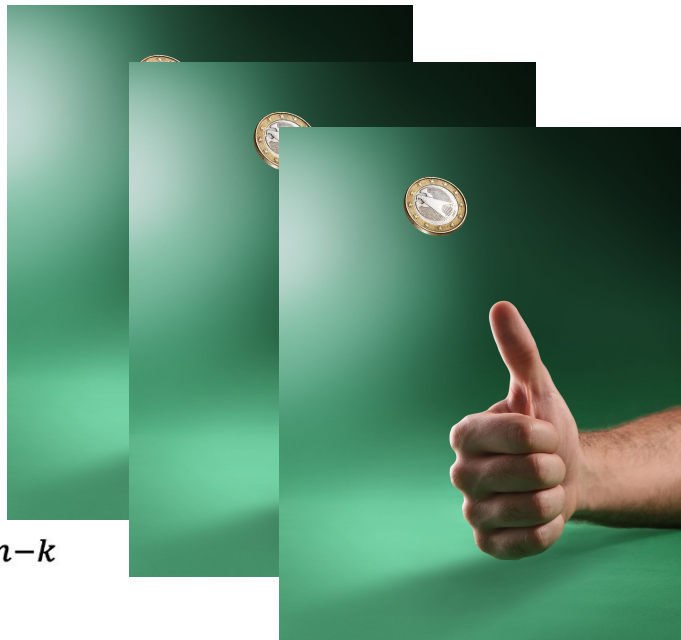


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- Examples:
 - Bernoulli Distribution:
 - Binomial Distribution:

$$\begin{cases} 1 - p & \text{for } x = 0 \\ p & \text{for } x = 1 \end{cases}$$

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

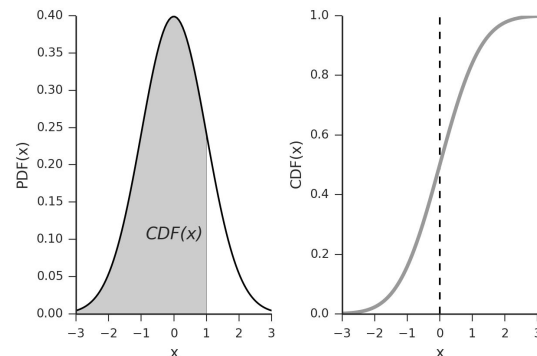


Continuous Probability Distribution

- A continuous random variable X is defined on a continuous sample space: an interval on the real line, a region in a high dimensional space, etc.
 - It is meaningless to talk about the probability of the random variable assuming a particular value --- $P(x) = 0$
 - Instead, we talk about the probability of the random variable assuming a value within a given interval, half interval, or arbitrary Boolean combination of basic propositions.

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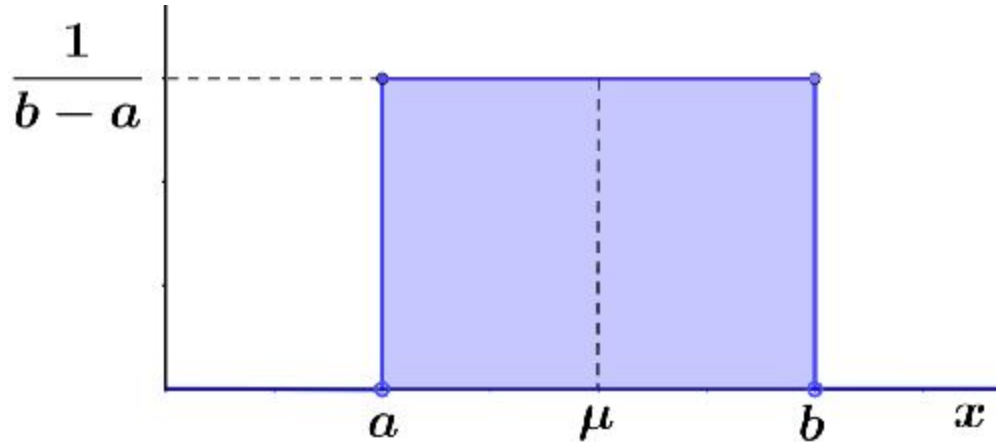


- Cumulative Distribution Function (CDF): $F_x(x) = P[X \leq x]$
- Probability Density Function (PDF): $F_x(x) = \int_{-\infty}^x f_x(x)dx$ or $f_x(x) = \frac{dF_x(x)}{dx}$
- Properties: $f_x(x) \geq 0$ and $\int_{-\infty}^{\infty} f_x(x)dx = 1$
- Interpretation: $f_x(x) = P \left[X \in \frac{x, x+\Delta}{\Delta} \right]$

Continuous Probability Distribution

- Examples:
 - Uniform Density Function:

$$f_x(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



Continuous Probability Distribution

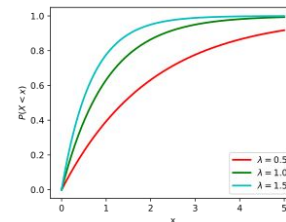
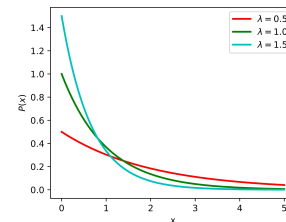
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 - Uniform Density Function:

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- Exponential Density Function:

$$f_x(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

$$F_x(x) = 1 - e^{-\lambda x} \quad \text{for } x \geq 0$$



Continuous Probability Distribution

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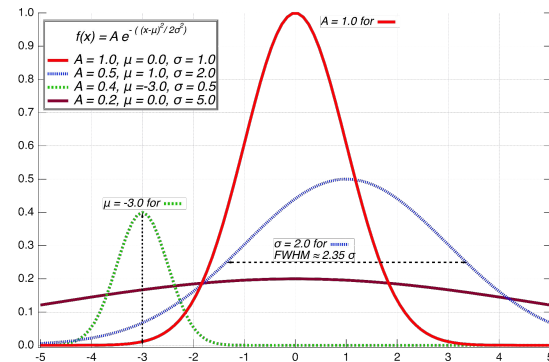
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- Gaussian(Normal) Density Function

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



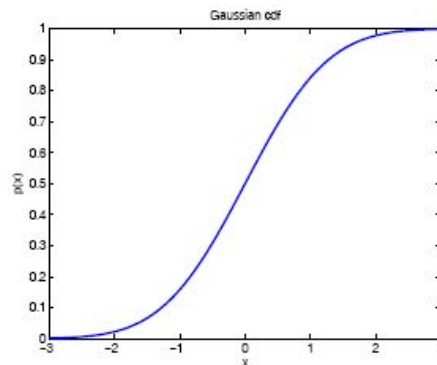
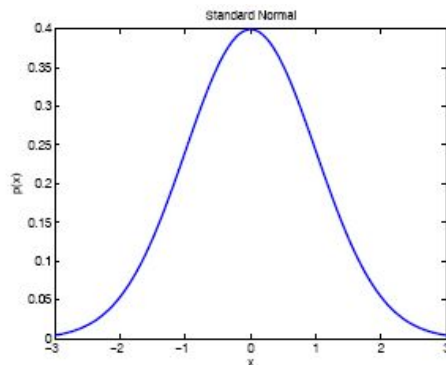
Continuous Probability Distribution

- Gaussian Distribution:

- If $Z \sim N(0,1)$

$$F_x(x) = \Phi(x) = \int_{-\infty}^x f_x(z) dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{z^2}{2}} dz$$

- This has no closed form expression, but is built in most software packages.



Statistics

- Expectation: The mean value, center of mass, first moment:

$$E_X[g(X)] = \int_{-\infty}^{\infty} g(x)p_X(x)dx$$

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$$E_X[g(X)] = \int_{-\infty}^{\infty} g(x)p_X(x)dx \quad \approx \quad \frac{1}{n} \sum_{i=1}^n g(x_i)$$

Statistics

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$$E_X[g(X)] = \int_{-\infty}^{\infty} g(x)p_X(x)dx$$

- N-th moment: $g(x) = x^n$
- N-th central moment: $g(x) = (x - \mu)^n$ $\mu = E_X[x]$

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- Mean: $E_X[X] = \int_{-\infty}^{\infty} xp_X(x)dx$
 - $E[\alpha X] = \alpha E[X]$
 - $E[\alpha + X] = \alpha + E[X]$
- Variance(Second central moment): $Var(x) = E_X[(X - E_X[X])^2] = E_X[X^2] - E_X[X]^2$
 - $Var(\alpha X) = \alpha^2 Var(X)$
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Statistics

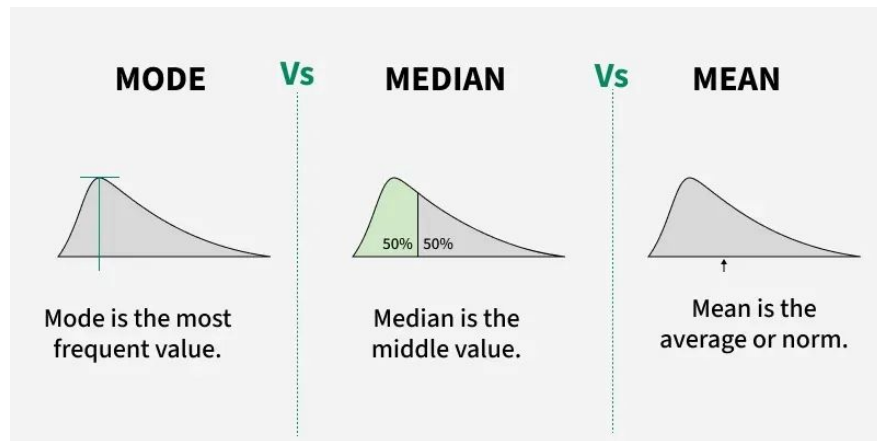
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Central Limit Theorem

- If $(X_1, X_2, \dots, X_n)$ are i.i.d. continuous random variables, then the joint distribution is $f(\bar{X})$
- CLT proves that $f(\bar{X})$ is Gaussian with mean $E[X_i]$ and $Var[X_i]$

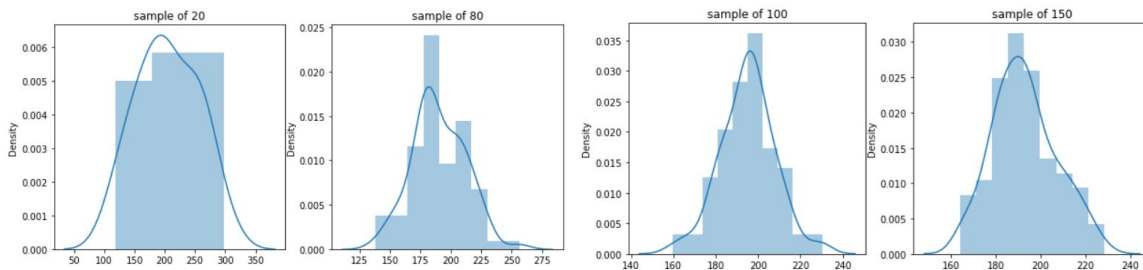
$$\bar{X} = f(X_1, X_2, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{as } n \rightarrow \infty$$

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- Somewhat of a justification for assuming Gaussian noise



Joint RVs and Marginal Densities

- Joint cumulative distribution:

$$F_{X,Y}(x, y) = P[\{X \leq x\} \cap \{Y \leq y\}] = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(\alpha, \beta) d\alpha d\beta$$

- Marginal densities:

- $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, \beta) d\beta$
- $p_X(x_i) = \sum_j p_{X,Y}(x_i, y_j)$

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- Expectation and Covariance:

- $E[X + Y] = E[X] + E[Y]$

- $cov(X, Y) = E[(X - E_X[X])(Y - E_Y[Y])] = E[XY] - E[X]E[Y]$

- $Var(X + Y) = Var(X) + 2cov(X, Y) + Var(Y)$

Conditional Probability

- $P(X | Y)$ = Fraction of the worlds in which X is true given that Y is also true.
- For example:
 - H = "Having a headache"
 - F = "Coming down with flu"
 - $P(\textit{Headche} | \textit{Flu})$ = fraction of flu-inflicted worlds in which you have a headache. How to calculate?
- Definition:

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This is called **Bayes Rule**

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$$P(\textit{Headache} | \textit{Flu}) = \frac{P(\textit{Headache}, \textit{Flu})}{P(\textit{Flu})} = \frac{P(\textit{Flu} | \textit{Headache})P(\textit{Headache})}{P(\textit{Flu})}$$

The Bayes Rule

$$P(\text{Headache} | \text{Flu}) = \frac{P(\text{Headache}, \text{Flu})}{P(\text{Flu})} = \frac{P(\text{Flu} | \text{Headache})P(\text{Headache})}{P(\text{Flu})}$$

- Other cases:

- $$P(Y | X) = \frac{P(X|Y)P(Y)}{P(X|Y)P(Y)+P(X|\neg Y)P(\neg Y)}$$

- $$P(Y = y_i | X) = \frac{P(X|Y)P(Y)}{\sum_{i \in S} P(X|Y=y_i)P(Y=y_i)}$$

- $$P(Y | X, Z) = \frac{P(X|Y,Z)P(Y,Z)}{P(X,Z)} = \frac{P(X|Y,Z)P(Y,Z)}{P(X|Y,Z)P(Y,Z)+P(X|\neg Y,Z)P(\neg Y,Z)}$$

Rules of Independence

- Recall that for events E and H , the probability of E given H , written as $P(E | H)$, is

$$P(E | H) = \frac{P(E, H)}{P(H)}$$

- E and H are (statistically) independent if

$$P(E, H) = P(E)P(H)$$

- Or equivalently

$$P(E) = P(E | H)$$

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- E and F are conditionally independent given H if

$$P(E | H, F) = P(E | H)$$

- Or equivalently

$$P(E, F | H) = P(E | H)P(F | H)$$

Suppose random variables Y, x and ϵ are related by $Y = \beta_0 + \beta_1 x + \epsilon$, with β_0 and β_1 are parameters and ϵ is assumed to independent of x and follow normal distribution with mean 0 and constant variance. Please calculate: (1) $E(\epsilon|x)$, and (2) $E(Y|x)$.

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$$E[\epsilon|x] = E[\epsilon] = 0$$

$$E[Y|x] = E[\beta_0 + \beta_1 x + \epsilon|x] = \beta_0 + \beta_1 x + E[\epsilon|x] = \beta_0 + \beta_1 x$$

- $E[X + Y] = E[X] + E[Y]$

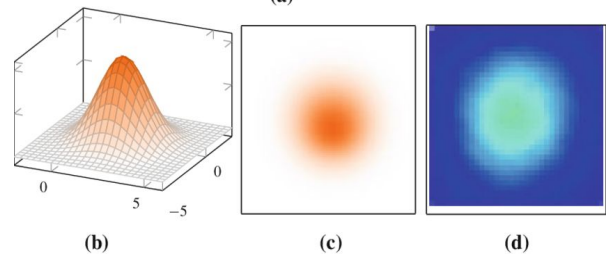
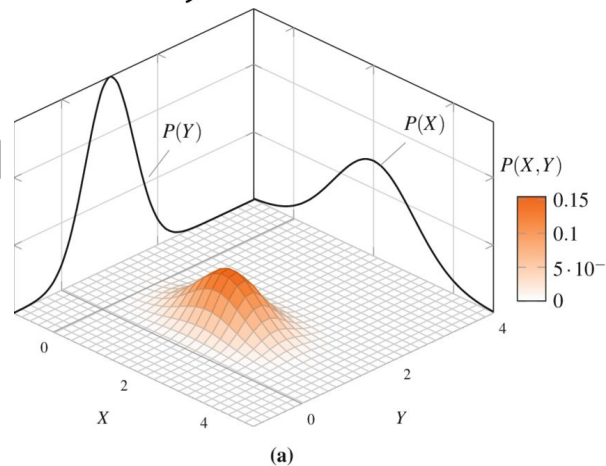
Multivariate Gaussian

$$p(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right\}$$

- Moment Parameterization $\mu = E(X)$

$$\Sigma = \text{Cov}(X) = E[(X - \mu)(X - \mu)^\top]$$

- Mahalanobis Distance $\Delta^2 = (x - \mu)^\top \Sigma^{-1} (x - \mu)$
- Tons of applications (MoG, FA, PPCA, Kalman filter,...)



Multivariate Gaussian

- Joint Gaussian $P(X_1, X_2)$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

- Marginal Gaussian

$$\mu_2^m = \mu_2 \quad \Sigma_2^m = \Sigma_2$$

- Conditional Gaussian $P(X_1 \mid X_2 = x_2)$

$$\begin{aligned} \mu_{1|2} &= \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2) \\ \Sigma_{1|2} &= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} \end{aligned}$$

Operations on Gaussian R.V.

- The **linear transform** of a Gaussian r.v. is a Gaussian. Remember that no matter how x is distributed

$$E(AX + b) = AE(X) + b$$

$$Cov(AX + b) = ACov(X)A^T$$

this means that for Gaussian distributed quantities:

$$X \sim N(\mu, \Sigma) \rightarrow AX + b \sim N(A\mu + b, A\Sigma A^T)$$

- The **sum** of two independent Gaussian r.v. is a Gaussian

$$Y = X_1 + X_2, X_1 \perp X_2 \rightarrow \mu_y = \mu_1 + \mu_2, \Sigma_y = \Sigma_1 + \Sigma_2$$

- The **multiplication** of two Gaussian functions is another Gaussian function (although no longer normalized)

$$N(a, A)N(b, B) \propto N(c, C)$$

$$\text{where } C = (A^{-1} + B^{-1})^{-1}, c = CA^{-1}a + CB^{-1}b$$

Maximum Log-Likelihood Estimation (MLE)

- Example: toss a coin
- Objective function:

$$l(\theta; \text{Head}) = \log P(\text{Head} \mid \theta) = \log \theta^n (1 - \theta)^{N-n} = n \log \theta + (N - n) \log (1 - \theta)$$

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- We need to maximize this w.r.t. θ
- Take derivatives w.r.t. θ


$$\frac{dl}{d\theta} = \frac{n}{\theta} - \frac{N-n}{1-\theta} = 0$$
$$\hat{\theta}_{MLE} = \frac{n}{N}$$

Maximum Log-Likelihood Estimation (MLE)

Given iid samples from Gaussian $\{x_i\}_{i=1}^n$

$$p(x_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Likelihood

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

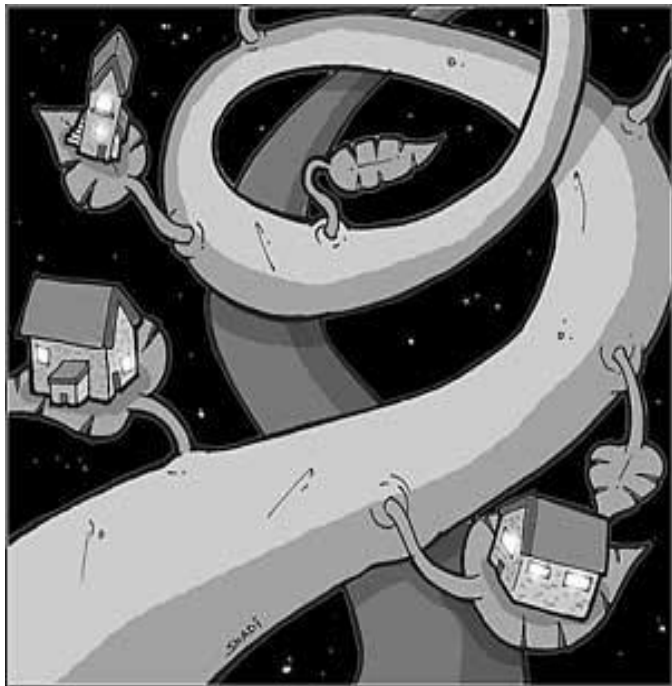
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$$\begin{aligned} \text{MLE} \quad \max_{\mu, \sigma} \ell(\mu, \sigma^2) &= \log L(\mu, \sigma^2) \\ &= -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \end{aligned}$$

Machine Learning for Apartment Hunting



- Suppose you are to move to Atlanta
- And you want to find the **most reasonably priced** apartment satisfying your **needs**:
monthly rent = $\theta_1(\text{living area}) + \theta_2(\# \text{ bedroom})$

Living area (ft ²)	# bedroom	Monthly rent (\$)
230	1	900
506	2	1800
433	2	1500
190	1	800
...		
150	1	?
270	1.5	?

Linear Regression Model

- Assume y is a linear function of x (features) plus noise ϵ
monthly rent = θ_1 (living area) + θ_2 (# bedroom)
$$y = \theta_0 + \theta_1 x_1 + \cdots + \theta_n x_n + \epsilon$$

where ϵ is an error model as Gaussian $N(0, \sigma^2)$

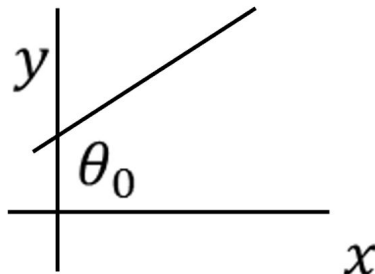
Probability

- Let $\theta = (\theta_0, \theta_1, \dots, \theta_n)^\top$, and augment data by one dimension

Linear algebra $x \leftarrow (1, x)^\top$

Then $y = \theta^\top x + \epsilon$

Linear algebra



Gaussian Likelihood

- Assume y is a linear in x plus noise ϵ

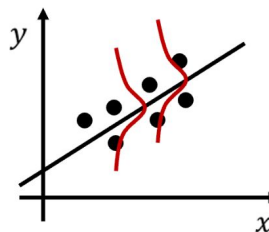
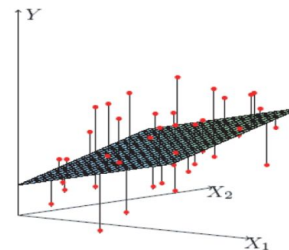
$$y = \theta^\top x + \epsilon$$

- Assume ϵ follows a Gaussian $N(0, \sigma)$

$$p(y^i | x^i; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^i - \theta^\top x^i)^2}{2\sigma^2}\right)$$

- By independence assumption, likelihood is

$$L(\theta) = \prod_i^m p(y^i | x^i; \theta) = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^m \exp\left(-\frac{\sum_i^m (y^i - \theta^\top x^i)^2}{2\sigma^2}\right)$$



Probability

MLE

$$L(\theta) = \prod_{i=1}^m p(y^i | x^i; \theta) = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^m \exp \left(- \frac{\sum_i^m (y^i - \theta^\top x^i)^2}{2\sigma^2} \right)$$

MLE

$$L(\theta) = \prod_{i=1}^m p(y^i | x^i; \theta) = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^m \exp \left(- \frac{\sum_{i=1}^m (y^i - \theta^\top x^i)^2}{2\sigma^2} \right)$$

$$\max_{\theta} \log L(\theta) = - \frac{1}{2\sigma^2} \sum_{i=1}^m (y^i - \theta^\top x^i)^2 - m \log(\sqrt{2\pi}\sigma)$$

Least Mean Square

Reference

- Chapter 2 in [Pattern Recognition and Machine Learning](#). Springer. 2006

Q&A